

Global Disaster Hotspot Mapping and Country-Level Vulnerability Analysis (2000–2025)

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1. Objective

The objective of this mini-project is to analyze global disaster patterns from 2000 to 2025 using real-world data from the EM-DAT International Disaster Database. Disasters continue to affect millions of people worldwide every year, and understanding their patterns is essential for humanitarian preparedness, policy-making, and risk reduction.

This project aims to achieve two major goals:

Identify global disaster hotspots using spatial analysis and mapping techniques.

Develop a country-level Vulnerability Index that quantifies the severity of disaster impacts by combining human losses, affected population, and economic damage.

In the process, the project applies core data science techniques including data cleaning, missing value handling, domain validation, outlier capping and statistical visualization using R.

2. Preliminary Analysis

The preliminary phase involved preparing the raw dataset for meaningful analysis. Five regional datasets—Africa, America, Asia, Europe, and Oceania—were merged into a comprehensive global dataset containing more than 16,000 disaster records.

2.1 Data Collection and Integration

The datasets were accessed directly from Google Drive and imported into R.

All regional files were combined into a single global dataset using row binding.

The merged dataset included various disaster characteristics such as date, location, type, deaths, affected population, and economic damage.

2.2 Data Cleaning

Several inconsistencies were addressed:

Numeric variables such as TotalDeaths, TotalAffected, and TotalDamage were converted from text formats into numeric formats.

Date fields were standardized by reconstructing complete start dates from the year, month, and day.

The dataset was filtered to include events occurring only between 2000 and 2025.

2.3 Handling Missing Data

Since real-world disaster data often contains missing values:

Missing Start Month and Start Day were imputed with safe defaults (Month = 1, Day = 1).

Missing deaths, affected population, or damage were kept as NA for analysis but replaced with zero only for specific visualizations requiring complete values.

Missing latitude/longitude values were excluded only from hotspot mapping, since they cannot be plotted.

2.4 Domain-Based Validation

To ensure the reliability of the dataset:

Latitude values outside -90 to 90 and longitude values outside -180 to 180 were removed.

Negative values for deaths, affected population, or damage were corrected to NA.

Invalid dates or corrupted entries were removed.

These validation steps ensured that the dataset adhered to real-world constraints.

2.5 Outlier Capping

Extreme disaster values can dominate global comparisons. To reduce skewness:

TotalDeaths, TotalAffected, and TotalDamage were capped using the 1st and 99th percentiles.

This ensured that rare mega-disasters did not distort the Vulnerability Index.

These preprocessing steps produced a stable, clean dataset for further analysis.

3. Detailed Analysis

Once the data was cleaned and validated, detailed analysis was carried out to understand spatial, temporal, and categorical patterns in global disasters.

3.1 Hotspot Mapping

Using the cleaned dataset, disaster events with valid coordinates were plotted on an interactive map:

Each event appeared as a circular marker, with size scaled according to total deaths.

Colors represented different disaster types such as floods, earthquakes, storms, and droughts.

Clustered markers made it easier to view patterns in regions with dense disaster histories.

This visualization clearly highlighted that Asia—particularly South and Southeast Asia—has been highly prone to major disasters in the last 25 years.

3.2 Construction of the Vulnerability Index

To quantitatively compare how severely disasters impact countries, a weighted index was defined. It combined three major components:

Total Deaths (50% weight) – representing human loss

Total Affected (30% weight) – showing the scale of humanitarian impact

Total Damage (20% weight) – capturing economic consequences

All components were scaled between 0 and 1 before weighting. Countries were then ranked according to this composite score.

This index helped identify countries that consistently suffer high human and economic impacts from disasters and are therefore more vulnerable.

3.3 Statistical Visualizations

Several visual patterns were studied:

A distribution of deaths showed a long-tailed pattern, where a few large events cause extremely high fatalities.

A bar chart of disaster types revealed that floods and storms are among the most frequently occurring disasters.

A period-wise regional comparison (2000–2010 vs. 2011–2025) showed increasing reporting and frequency of disasters.

A year-wise event frequency plot showed fluctuations but an overall rising trend, especially in climate-related events.

These visual insights helped strengthen the understanding of how disasters evolve over time and vary across regions.

4. Outcomes

The project resulted in a comprehensive understanding of global disaster trends. The key outcomes include:

A fully cleaned and validated disaster database covering 25 years.

An interactive geospatial hotspot map illustrating global disaster concentration.

A country-level Vulnerability Index that numerically reflects a country's overall disaster severity and exposure.

Multiple exploratory visualizations offering insights into annual trends, regional distributions, and major disaster types.

A deeper understanding of data preprocessing challenges such as missing values, outliers, and dataset inconsistencies.

Overall, the project demonstrates the practical application of R in real-world data analysis tasks involving uncertainty and complexity.

5. Results & Discussion

5.1 Regional Insights

Asia emerges as the most disaster-prone region due to high frequencies of earthquakes, floods, and storms. Africa is significantly affected by slow-onset disasters like droughts, impacting millions over long periods. The Americas show high economic losses despite fewer deaths, reflecting better preparedness and reporting.

5.2 Vulnerability Insights

Countries such as Bangladesh, Nepal, and several African nations rank highly vulnerable due to a mix of high exposure, high population density, and limited resilience.

High-income countries show high economic losses but stronger disaster management, resulting in lower vulnerability scores.

5.3 Data Quality Observations

Missing location data reduced the number of events available for mapping.

Outlier capping improved fairness in country comparisons.

Domain validation prevented unrealistic values from skewing results.

5.4 Overall Interpretation

The combination of geospatial mapping and vulnerability indexing provides a holistic understanding of global disaster impacts. The results emphasize that disaster risk is not determined solely by frequency, but also by population, resilience, infrastructure, and national preparedness.